# Real-Time Crowd Sound Generation from Chatroom Messages via Affective Analysis


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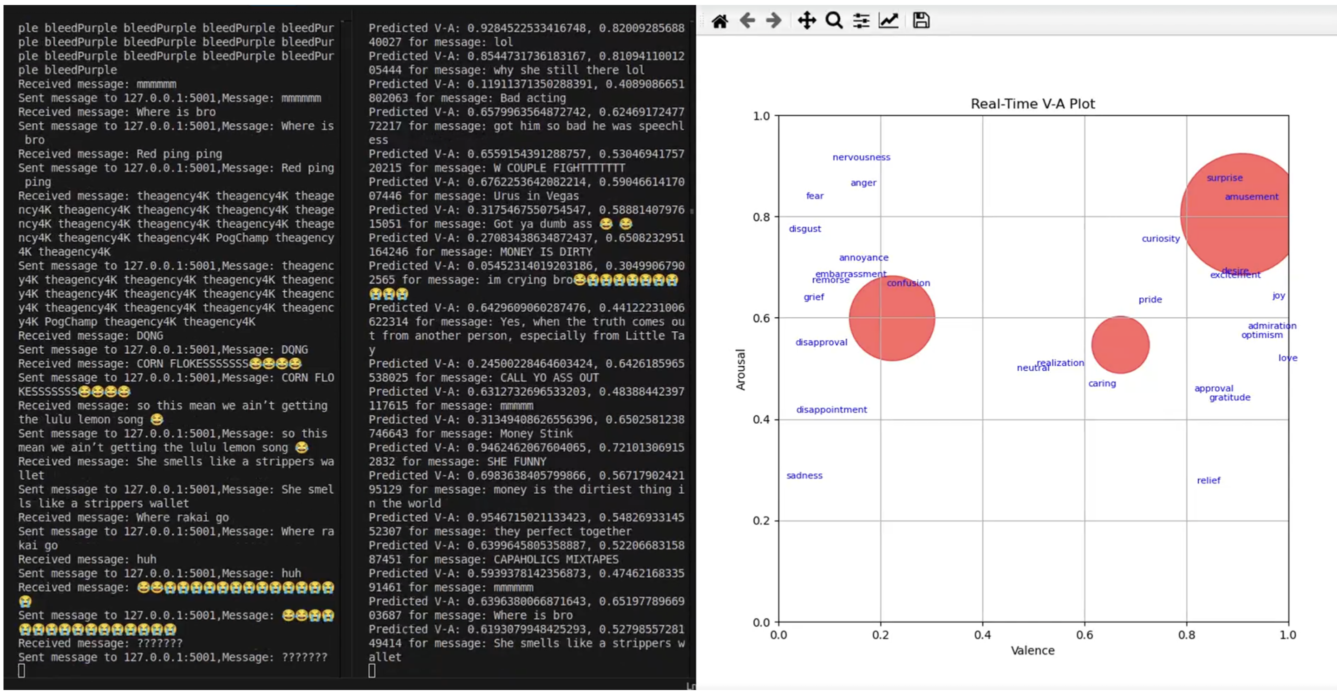


**Figure 1: Screenshot of the chat to valence/arousal mapping, with visualization.**


## Abstract

Text-based chatrooms on live streaming platforms generate a rapid stream of emotional content that remains visually dense and difficult to parse in real time. This hinders streamers and viewers from understanding the affective dynamics of their audience. We propose a system that sonifies live chat message streams by converting chatroom messages into expressive sounds using real-time text sentiment analysis. Our method involves three stages: fine-tuning BERT for valence-arousal regression, clustering emotional states, and mapping those states to audio using the Emo-Soundscapes dataset and a Python-based audio program. Our contribution provides a novel approach to social signal amplification, bridging NLP and audio synthesis to enrich digital audience interaction.



## Keywords

Affective computing, Social signal processing, Real-time systems, Emotion sonification, Text-to-audio, Chatroom analysis

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## 1 Introduction

Text-based chatrooms on live streaming platforms are dynamic spaces where emotions emerge, shift, and cascade across large groups. Yet, these emotional waves are typically trapped within fast-scrolling text which are difficult to interpret in real time and easy to overlook. While the streams of text are often rich with affective expression, they remain impossible to parse at scale by hosting streamer, making it difficult for the streamer to react and respond to their audience in real-time.

We explore a new modality: text-based crowd emotion sonification. By analyzing the emotional dynamics of live chat messages and generating expressive crowd sound, we aim to make an audience's emotion not just visible but audible too. Our goal is to design a system that transforms live chatroom messages into affective crowd soundscapes, allowing listeners to hear the mood and energy-level of the crowd in real time.

Our work draws inspiration from high-impact data sonification projects that translate complex phenomena into immersive audio experiences, such as NASA's sonification of a black hole [1]. Prior research in affective computing and social signal processing has made significant progress in understanding emotion from text, yet few works explore translating these signals into real-time modalities that can be *heard* rather than read. While datasets like GoEmotions and NRC VAD lexicon have enabled prior emotion classification systems, our work is novel in its real-time use of these emotional signals to drive audio synthesis, thereby creating an affective interface for digital crowd dynamics [2, 7]. Our work is also inspired by the affective crowd sound classifiers and datasets based on recordings from sports events [3]. This contribution lies at the intersection of natural language processing, affective computing, and audio interaction design, offering a new way to experience collective audience emotion.

## 2 Approach

Our **text-to-sound generation pipeline** consists of three key stages:

(1) **Text-to-affect regression**
(2) **Affect aggregation**
(3) **Affect-to-audio mapping**

In the **text-to-affect regression** step, live chat messages are retrieved via an API. Each message is processed using a fine-tuned BERT model to extract a corresponding valence-arousal pair.

Next, during **affect aggregation**, the system collects a series of valence-arousal pairs over a short time window. These are then clustered to identify the top three most representative affective states.

Finally, in the **affect-to-audio mapping** step, the dominant valence-arousal pairs are mapped to audio clips that reflect the corresponding emotional tone. The **loudness** of each audio output is scaled based on the size of its associated cluster.

### 2.1 Step 1: Text-to-Affect Regression

To obtain valence-arousal pairs not only for a preset collection of words but also for open-ended sentences—such as those commonly found in live chat streams—the team fine-tuned a language model using manually labeled online comment data.

GoEmotions is a corpus of 58,000 carefully curated comments extracted from Reddit, each annotated by humans with one of 27 emotion categories or Neutral [4]. These 28 labels were mapped to valence-arousal pairs using the NRC Valence-Arousal-Dominance (VAD) lexicon [7]. This transformation preserves the relative proximity between emotions in affective space. For example, a positive emotion such as *Amusement* is closer to other positive emotions like *Approval* than to negative emotions such as *Fear*.

**BERT** [5, 9] was selected due to its strong performance on contextual language understanding and its proven effectiveness in emotion classification tasks. A BERT-base model was fine-tuned using the GoEmotions dataset, with the labels replaced by valence-arousal pairs as described above.

*2.1.1 Training Setup.* We fine-tuned a BERT-base model using the Hugging Face Transformers library. The training was conducted on a single NVIDIA RTX 4090 GPU. The training configuration is as follows:

- **Learning rate:** 1e-4
- **Batch size:** 32 (for both training and evaluation)
- **Number of epochs:** 8
- **Weight decay:** 0.01

*2.1.2 Loss Function and Evaluation Metrics.* The model is trained as a regression task, predicting a pair of valence and arousal values for each input. We use the Mean Squared Error (MSE) as the loss function.

During evaluation, we also compute an auxiliary classification accuracy. Each predicted and ground truth valence-arousal pair is matched to the closest predefined emotion label by minimizing the Euclidean distance in affective space. Let $P \in \mathbb{R}^{N\times 2}$ be the predicted valence-arousal vectors, and $T \in \mathbb{R}^{N\times 2}$ be the ground truth labels. The nearest emotion class is assigned by:

$$\text{label}_i = \arg\min_j \|x_i - v_j\|_2$$

where $x_i$ is the predicted or true valence-arousal pair, and $v_j$ is the valence-arousal pair of the $j$-th emotion in the predefined emotion set.

| **Emotion** | **Valence** | **Arousal** |
|---|---|---|
| Admiration | 0.969 | 0.583 |
| Amusement | 0.929 | 0.837 |
| Anger | 0.167 | 0.865 |
| Annoyance | 0.167 | 0.718 |
| ... | ... | ... |

**Table 1: Valence and arousal scores for each emotion label.(Full table in Appendix)**

*2.1.3 Training Loss.* The training loss plot is included in the Appendix.

### 2.2 Step 2: Affect Aggregation

We applied the Elbow method on 10 sets of sample data, each containing approximately 100 entries, testing $k$ values ranging from 2 to 15. The results indicated that $k = 3$ generally produced the most optimal clustering outcome.

During inference, the clustering process is updated every 30 seconds. This time interval balances the need to gather a sufficient number of data points with the goal of maintaining a smooth and non-disruptive background audio experience. According to our measurements, a stream with 10,000 viewers typically generates around 300 chat messages every 30 seconds.

To reduce noise and eliminate semantically empty chat entries, we filtered out all valence-arousal pairs close to neutral. Specifically, any entry with an Euclidean distance from the origin (0.5,0.5) less than 0.15 was discarded, as it had negligible effect on the generated audio output.

K-Means clustering is then applied to the remaining valence-arousal points to identify the dominant affective states within each time window. For each resulting cluster, we extract the following:

- The **centroid** (mean valence and arousal), representing the affective center of the cluster.
- The **intensity**, defined as the relative proportion of messages assigned to the cluster.

These values—*(valence, arousal, intensity)* triples—form the emotional control signals used for real-time sound generation.

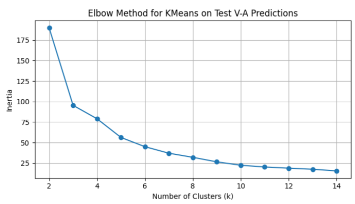


**Figure 2: Elbow Method Plot**

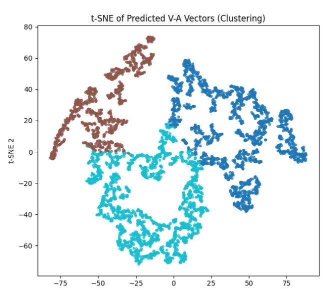


**Figure 3: Cluster Result Plot**

## 2.3 Step 3: Affect-to-Audio Mapping

We leverage the Emo-Soundscapes dataset, which provides 1213 six-second Creative Commons–licensed audio clips annotated with perceived valence and arousal through crowdsourcing perceived rankings. We designed a real-time mixing system in Python using sounddevice and soundfile that maps incoming valence/arousal/intensity signals to the closest ranked sound clips from the Emo-Soundscapes dataset. Our approach dynamically selects up to N “closest” sound clips by comparing each incoming signal’s valence and arousal scores to those in Emo-Soundscapes, while also factoring in a user-defined randomness threshold. We scale the overall volume and number of layered clips based on the intensity dimension, ensuring that a higher-intensity signal yields louder playback and more clips.

# 3 Datasets

To support affective modeling and real-time sound synthesis, we incorporate three publicly available datasets spanning both textual and auditory modalities: GoEmotions, NRC Valence–Arousal–Dominance (VAD) Lexicon, and Emo-Soundscapes.

GoEmotions [2] is a large-scale corpus of 58,000 English Reddit comments manually annotated with 28 emotion categories—27 fine-grained emotions and one neutral class. Each sample is labeled by multiple annotators in a multi-label setting, providing a diverse and nuanced representation of everyday emotional expression in online discourse. In our work, we fine-tune a BERT-based regression model using GoEmotions, repurposing the original categorical labels as input to a continuous valence–arousal mapping.

The NRC-VAD lexicon [8] contains human-annotated affective ratings for over 20,000 English words. Each word is assigned a numerical score in three continuous dimensions: valence, arousal, and dominance. We use this lexicon to assign each GoEmotions category a canonical valence–arousal pair by matching the label term (e.g., joy, anger) to its corresponding entry in the lexicon. These (valence, arousal) values are then used as regression targets in our social-affective signal estimation model.

Emo-Soundscapes [6] is a Creative Commons–licensed dataset of 1,213 six-second audio clips, each annotated with perceived valence and arousal values via crowdsourced pairwise comparisons. The clips span a wide range of ambient and synthetic textures designed to evoke varying affective impressions. In our system, we use Emo-Soundscapes as a retrieval base for affect-conditioned crowd sound synthesis. Valence–arousal predictions are mapped to the nearest clips in this dataset to generate dynamic and emotionally resonant audio feedback.

# 4 Experiments and Results

Our system generates expressive, real-time audio summaries of live chatrooms based on social signal modeling. It processes batches of messages, projects them into a continuous valence–arousal space, and synthesizes crowd sound accordingly.

Figure 1 presents an example of raw valence–arousal predictions for individual chat messages collected during a 30-second window from the Twitch API. Figure 1 visualizes the aggregated affective state of the same message batch by clustering the predicted values and plotting their centroids in the valence–arousal space.

## 4.1 Evaluation Metrics

Model performance was evaluated using:

- **Mean Squared Error (MSE)** between predicted and true VA vectors.
- **Valence-Arousal Classification Accuracy**, computed by mapping each VA pair to its nearest discrete emotion label using Euclidean distance.

## 4.2 Classification Performance

Given the ambiguous and layered nature of emotional language in live chat, our system benefits from **approximate affective correctness** rather than strict categorical precision. Therefore, we evaluate both Top-1 and Top-5 performance. A correct Top-5 prediction is counted if the true label is among the model's top five candidates.

| Metric | Top-1 | Top-5 |
|---|---|---|
| Accuracy | 39.75% | 64.47% |
| Precision (weighted) | 47.81% | 67.87% |
| Recall (weighted) | 39.75% | 64.47% |
| F1 Score (weighted) | 42.76% | 65.26% |

**Table 2: Comparison of Top-1 and Top-5 classification performance. Baseline Top-1 accuracy for 28 classes is approximately 3.57%.**

### 4.3 Confusion Matrix

To visualize the broader classification behavior in Top-1 and Top-5 setting, we generated a confusion matrix using pseudo-label matching. The confusion matrix is included in the Appendix.

### 4.4 High-Performing Categories

Table 3 highlights categories with strong performance under both Top-1 and Top-5 evaluations.

| Emotion Category | Top-1 Accuracy | Top-5 Accuracy |
|---|---|---|
| Gratitude | 74.77% | 90.97% |
| Love | 68.72% | 85.90% |
| Admiration | 30.75% | 76.44% |
| Joy | 26.67% | 74.07% |
| Remorse | 15.09% | 83.02% |

**Table 3: High-performing emotion classes under Top-1 and Top-5 evaluation.**

### 4.5 Full Classification Reports

Complete per-class precision, recall, and F1-score metrics for both Top-1 and Top-5 settings are included in Appendix D.

### 4.6 Qualitative Examples

To validate the model's interpretability, several chat messages were manually tested. For example:

- `"Let's goooo!!"` → Predicted: (0.91, 0.88), Closest Emotion: *Excitement*
- `"That was a huge mistake..."` → Predicted: (0.15, 0.60), Closest Emotion: *Disappointment*

The predictions show semantic alignment between user expressions and the underlying affective state.

## 5 Discussion

Our system offers a novel modality for rendering real-time crowd affect through audio, and several aspects of its behavior warrant further reflection.

One of the most promising outcomes was the system's ability to continuously generate affect-informed soundscapes that subjectively matched the overall tone of chatroom activity. Informal feedback from several listeners, both technical and non-technical, suggested that the audio output enhanced their sense of presence and offered a "feeling of being with the crowd" even when watching streams asynchronously. These anecdotal reactions hint at the potential of ambient sonification as a lightweight but emotionally resonant channel for live audience awareness.

Beyond subjective impressions, the system also demonstrated strong modeling performance in quantitative evaluations. Our fine-tuned BERT model achieved over 64% Top-5 classification accuracy across 28 emotion categories, far exceeding the baseline and confirming its ability to interpret a diverse range of emotional signals.

The system was also shown to be capable of use in real time, with a low-latency pipeline integrating message processing, clustering, and audio mixing without blocking execution. Our audio synthesis design modulated both the number and volume of layered clips and effectively conveyed changes in crowd "intensity," even in short intervals. Finally, the approach is platform-agnostic and can be adapted to any live textual input stream, including gaming chats, virtual events, or online classrooms.

One limitation of our system is that the fine-tuned BERT model struggled with certain low-frequency or subtle emotion categories such as relief, embarrassment, and grief. This is likely due to both class imbalance in the GoEmotions dataset and the challenges of mapping nuanced expressions to a 2D affective space.

Looking ahead, we identify several directions for improvement. First, a formal user study is needed to assess the emotional interpretability and impact of the generated audio. Second, we plan to explore alternative emotion mapping models that consider chat history, emoji use, and message timing. Finally, more expressive audio synthesis techniques, such as generative diffusion models, may allow for more fine-grained, reactive soundscapes that reflect both the emotion and rhythm of live chat.

## 6 Conclusion

We presented a real-time system that transforms live chatroom messages into affect-informed crowd audio using a pipeline of social-affective signal modeling and audio synthesis. By combining valence–arousal regression, affective clustering, and emotion-aligned sound retrieval, our approach offers a new modality for perceiving audience sentiment in dynamic social environments. Preliminary results demonstrate both technical feasibility and promising user-facing potential. Future work could explore listener perception and adaptive synthesis methods to enhance expressiveness.

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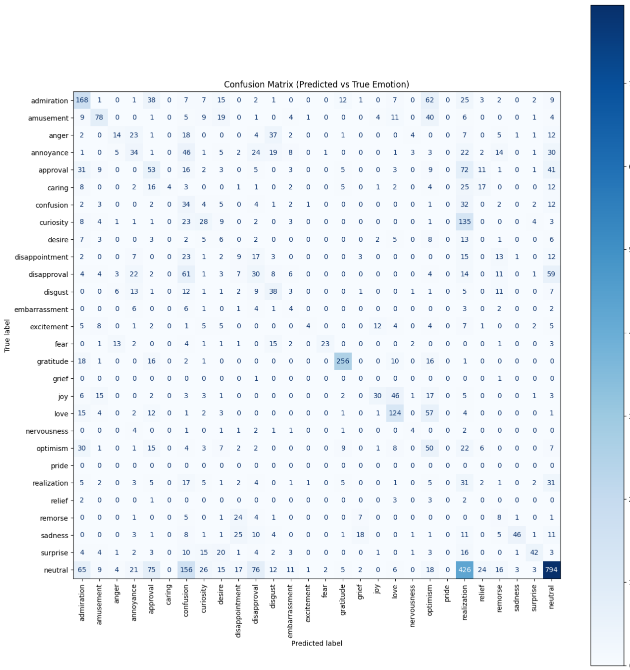
# A Valence and Arousal Mapping

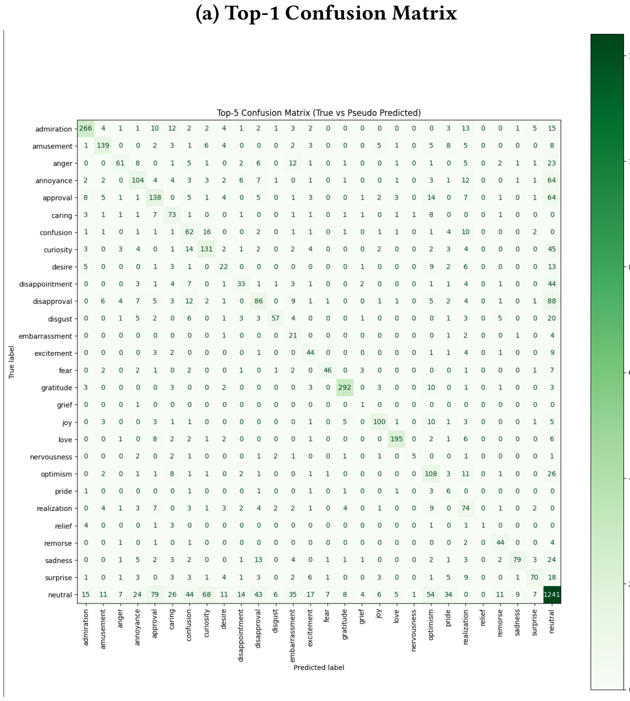
| Emotion | Valence | Arousal |
|---|---|---|
| Admiration | 0.969 | 0.583 |
| Amusement | 0.929 | 0.837 |
| Anger | 0.167 | 0.865 |
| Annoyance | 0.167 | 0.718 |
| Approval | 0.854 | 0.460 |
| Caring | 0.635 | 0.469 |
| Confusion | 0.255 | 0.667 |
| Curiosity | 0.750 | 0.755 |
| Desire | 0.896 | 0.692 |
| Disappointment | 0.104 | 0.418 |
| Disapproval | 0.085 | 0.551 |
| Disgust | 0.052 | 0.775 |
| Embarrassment | 0.143 | 0.685 |
| Excitement | 0.896 | 0.684 |
| Fear | 0.073 | 0.840 |
| Gratitude | 0.885 | 0.441 |
| Grief | 0.070 | 0.640 |
| Joy | 0.981 | 0.643 |
| Love | 1.000 | 0.519 |
| Nervousness | 0.163 | 0.915 |
| Optimism | 0.949 | 0.565 |
| Pride | 0.729 | 0.634 |
| Realization | 0.554 | 0.510 |
| Relief | 0.844 | 0.278 |
| Remorse | 0.103 | 0.673 |
| Sadness | 0.052 | 0.288 |
| Surprise | 0.875 | 0.875 |
| Neutral | 0.500 | 0.500 |

**Table 4: Valence and arousal scores for each emotion label.**

# B Confusion Matrices




**(a) Top-1 Confusion Matrix**




**(b) Top-5 Confusion Matrix**

**Figure 4: Side-by-side comparison of Top-1 and Top-5 confusion matrices.**

## C Top-1 Classification Report

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| admiration | 0.48 | 0.31 | 0.37 | 348 |
| amusement | 0.64 | 0.58 | 0.60 | 193 |
| anger | 0.44 | 0.29 | 0.35 | 132 |
| annoyance | 0.23 | 0.15 | 0.18 | 223 |
| approval | 0.24 | 0.27 | 0.25 | 265 |
| caring | 0.09 | 0.18 | 0.12 | 103 |
| confusion | 0.13 | 0.25 | 0.17 | 107 |
| curiosity | 0.39 | 0.35 | 0.37 | 223 |
| desire | 0.06 | 0.08 | 0.07 | 63 |
| disappointment | 0.15 | 0.13 | 0.14 | 108 |
| disapproval | 0.27 | 0.20 | 0.23 | 240 |
| disgust | 0.51 | 0.27 | 0.35 | 112 |
| embarrassment | 0.01 | 0.03 | 0.01 | 30 |
| excitement | 0.17 | 0.24 | 0.20 | 66 |
| fear | 0.59 | 0.48 | 0.53 | 69 |
| gratitude | 0.85 | 0.75 | 0.80 | 321 |
| grief | 0.00 | 0.00 | 0.00 | 2 |
| joy | 0.45 | 0.27 | 0.33 | 135 |
| love | 0.83 | 0.69 | 0.75 | 227 |
| nervousness | 0.14 | 0.06 | 0.08 | 18 |
| optimism | 0.14 | 0.30 | 0.19 | 168 |
| pride | 0.02 | 0.20 | 0.04 | 15 |
| realization | 0.03 | 0.10 | 0.05 | 124 |
| relief | 0.00 | 0.00 | 0.00 | 11 |
| remorse | 0.14 | 0.15 | 0.15 | 53 |
| sadness | 0.70 | 0.37 | 0.48 | 148 |
| surprise | 0.55 | 0.32 | 0.40 | 136 |
| neutral | 0.61 | 0.51 | 0.56 | 1787 |
| | | | | |
| accuracy | | | 0.40 | 5427 |
| macro avg | 0.32 | 0.27 | 0.28 | 5427 |
| weighted avg | 0.48 | 0.40 | 0.43 | 5427 |

## D Top-5 Classification Report

Top-5 Classification Report:

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| admiration | 0.85 | 0.76 | 0.80 | 348 |
| amusement | 0.77 | 0.72 | 0.75 | 193 |
| anger | 0.73 | 0.46 | 0.56 | 132 |
| annoyance | 0.59 | 0.47 | 0.52 | 223 |
| approval | 0.50 | 0.52 | 0.51 | 265 |
| caring | 0.46 | 0.71 | 0.56 | 103 |
| confusion | 0.34 | 0.58 | 0.43 | 107 |
| curiosity | 0.56 | 0.59 | 0.57 | 223 |
| desire | 0.34 | 0.35 | 0.35 | 63 |
| disappointment | 0.49 | 0.31 | 0.38 | 108 |
| disapproval | 0.48 | 0.36 | 0.41 | 240 |
| disgust | 0.80 | 0.51 | 0.62 | 112 |
| embarrassment | 0.20 | 0.70 | 0.31 | 30 |
| excitement | 0.47 | 0.67 | 0.55 | 66 |
| fear | 0.81 | 0.67 | 0.73 | 69 |
| gratitude | 0.92 | 0.91 | 0.92 | 321 |
| grief | 0.07 | 0.50 | 0.12 | 2 |
| joy | 0.80 | 0.74 | 0.77 | 135 |
| love | 0.92 | 0.86 | 0.89 | 227 |
| nervousness | 0.71 | 0.28 | 0.40 | 18 |
| optimism | 0.43 | 0.64 | 0.52 | 168 |
| pride | 0.08 | 0.40 | 0.13 | 15 |
| realization | 0.39 | 0.60 | 0.47 | 124 |
| relief | 1.00 | 0.09 | 0.17 | 11 |
| remorse | 0.61 | 0.83 | 0.70 | 53 |
| sadness | 0.85 | 0.53 | 0.66 | 148 |
| surprise | 0.74 | 0.51 | 0.61 | 136 |
| neutral | 0.72 | 0.69 | 0.71 | 1787 |
| | | | | |
| accuracy | | | 0.64 | 5427 |
| macro avg | 0.59 | 0.57 | 0.54 | 5427 |
| weighted avg | 0.68 | 0.64 | 0.65 | 5427 |

## E Training Loss

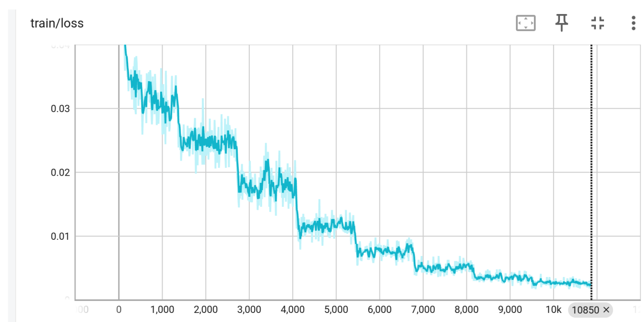



**Figure 5: Training loss**

## F Contribution

Qihan:

- Initiating group meetings - Model training and testing and visualizations - Architecture - Proof-reading and document contributions

Xueying:

- Document creation and design - Poster design, creation and printing

Sarah: - Affect to audio mapper programming - Brainstorming - Proof-reading and Document contributions